

ECE 647: Lecture 21

Fading + Multi-user

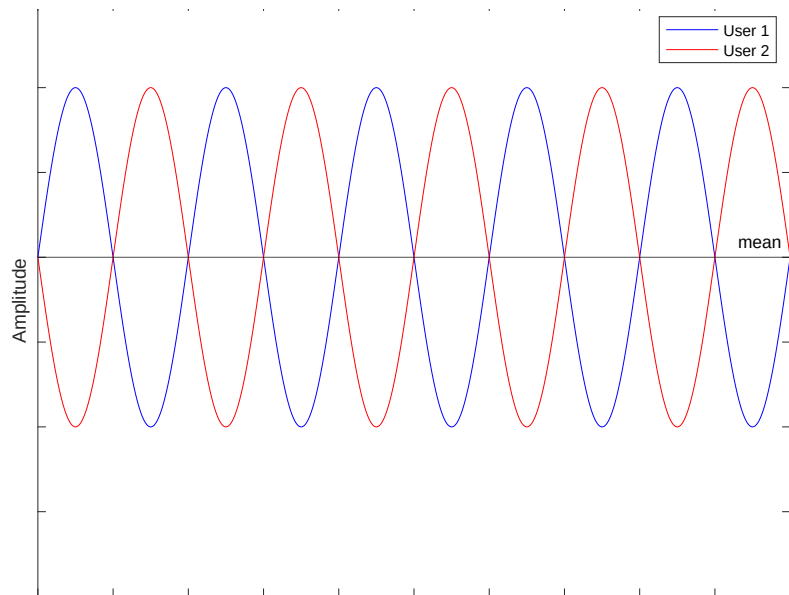
- The channel can be in one of K states.
- Assume two users for simplicity.
- If the system is at state k , and user i ($= 1, 2$) is chosen for transmission, the payoff to the chosen user is U_i^k . The payoff is zero for the other user.
- Power is otherwise fixed beforehand.

How to choose which user to transmit?

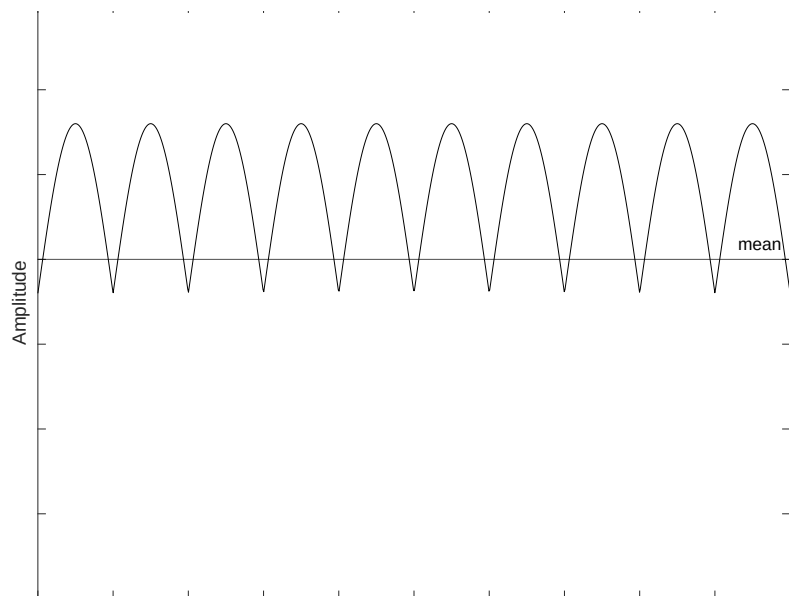
- If we choose each user uniformly randomly with probability $\frac{1}{2}$:

$$\frac{1}{2}E[U_i^k] + \frac{1}{2}E[U_i^k] \tag{1}$$

- Average performance of each user:



- If we instead choose the user with the better rate:



- Achieves higher overall performance.

- Improvement for both users.

Should we always pick the user with higher rate?



- The worse user could be starved.
- Need to consider the tradeoff between performance and fairness.

Channel Model

- Let $g_k = \Pr\{\text{channel is at state } k\}$.
- Power is fixed, so the rate is completely determined by the channel state.
- U_i^k : utility.
- $Q_i^k = \begin{cases} 1, & \text{if the policy decides that user 1 should be chosen at state } k, \\ 0, & \text{otherwise.} \end{cases}$
- The expected payoff to user i is:

$$\sum_k g_k Q_i^k U_i^k \quad (2)$$

- The system payoff is:

$$\sum_i \sum_k q_k Q_i^k U_i^k \quad (3)$$

- Need a way to balance the system payoff vs. individual payoff. This can be done by imposing a fairness constraint.

a) Temporal Fairness

- Fraction of time user i is scheduled:

$$\sum_k q_k Q_i^k \quad (4)$$

- Fairness constraint in time: user i should be scheduled at least r_i fraction of time:

$$\max \sum_i \sum_k q_k Q_i^k U_i^k \quad (5)$$

subject to:

$$\sum_k q_k Q_i^k U_i^k, \forall i \quad (6)$$

$$\sum_i Q_i^k = 1, \forall k \quad (7)$$

$$Q_i^k \in \{0, 1\} \quad (8)$$

- This is not a convex problem. To fix this, instead, relax the last constraint to be:

$$Q_i^k \in [0, 1] \quad (9)$$

where Q_i^k represents the probability that user i is chosen if the state is k .

- This is a linear problem.

Other types of fairness constraints:

- a) Temporal Constraint
- b) Utility Constraint
- c) Fraction of Utility Constraint
- etc.

Approximation

High SINR Approximation

$$g_l(\mathbf{p}) = \log \left(1 + \frac{p_l}{\sum_{j \neq l} p_j + n_l} \right) \approx \log \left(\frac{p_l}{\sum_{j \neq l} p_j + n_l} \right) \quad (10)$$

- For simplicity, assume single-path routing and no fading. This becomes:

$$\max \sum_s U_s(x_s) \quad (11)$$

subject to:

$$\sum_s H_s^l x_s \leq \log \left(\frac{p_l}{\sum_{j \neq l} p_j + n_l} \right), \quad l = 1, \dots, L \quad (12)$$

$$p_l \leq P_{\max} \quad (13)$$

- Through a change in variables: $p_l = e^{Q_l}$, This transforms into:

$$\max \sum_s U_s(x_s) \quad (14)$$

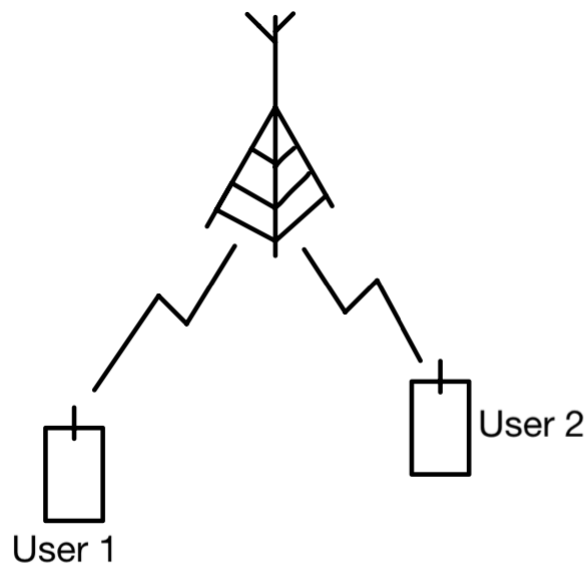
subject to:

$$\sum_s H_s^l x_s \leq Q_l - \log \left(\sum_{j \neq l} e^{Q_j} + N_l \right), \quad l = 1, \dots, L \quad (15)$$

Pitfalls of this approach:

- Solution may not lead to high SINR.
- As a result, the quality of the solution may be worse than scheduling or time-interleaving.

Example:



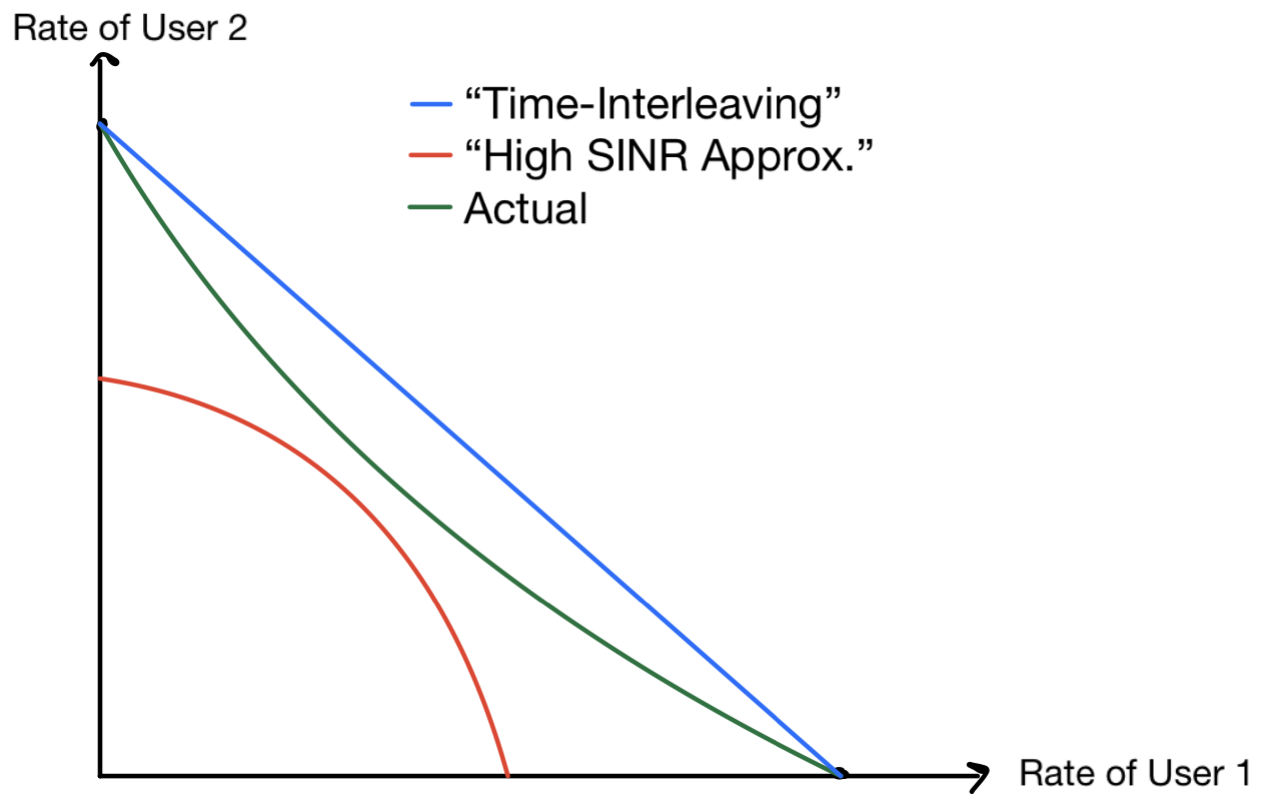
- **High SINR Approximation** - both users transmit at the same time with rate:

$$\log\left(1 + \frac{P}{P + N}\right) \quad (16)$$

- **Time-interleaving** - One user transmits at a time. Average rate:

$$\frac{1}{2} \log\left(1 + \frac{P}{N}\right) \quad (17)$$

- **When N is small, option (b) is often much better:**



Two lessons:

1. Convex approximations are not always better.
2. Need to use domain knowledge to pick the right formulation/approximation.